

HOLYFAITH ACADEMY, DHULIYAN

Affiliated to CISCE, New Delhi

Semester – II Examination

Session: 2025-26

Computer Applications [*Theory*]

Class: IX

FM: 100 marks

Section-A

Q 1.

[5 X 2 = 10]

- (a) What do you understand by the term class?
- (b) What is polymorphism?
- (c) What is byte code in Java programming language?
- (d) What are predefined methods? Give an example.
- (e) How is a class a user defined data type?

Q 2.

[5 X 1 = 5]

- (a) Which one is a valid declaration of a Boolean?
 - (i) Boolean b1 = 'true';
 - (ii) Boolean b2 = 'false';
 - (iii) Boolean b3 = 1;
 - (iv) boolean b4 = true;
- (b) What is the numerical range of a char in Java?
 - (i) -128 to 127
 - (ii) 0 to 256
 - (iii) 0 to 65535
 - (iv) 0 to 32767
- (c) Evaluate the value of the expression $6 - 2 + 10 \% 4 + 7$
 - (i) 10
 - (ii) 12
 - (iii) 14
 - (iv) 13
- (d) Which of this method of class String is used to extract a single character from a String object?
 - (i) CHARAT()
 - (ii) charAt()
 - (iii) charat()
 - (iv) CharA+()
- (e) Which of these are selection statements in Java?
 - (i) If
 - (ii) for
 - (iii) continue
 - (iv) break;

Q 3.

[5 X 1 = 5]

- (a) What is ternary operator?
- (b) What is a constructor?
- (c) What is the difference between parameters and argument.
- (d) Explain Shorthand Operators.
- (e) What are operands?

Q 4.

[2 X 10 = 20]

- (a) The output of `Math.max (-7, Math.min (-5, -4))` is:
 - (i) -5
 - (ii) -4
 - (iii) -7
 - (iv) error
- (b) Write the Java expression to find the product of square root of P and the square root of Q using the methods of Math class.
- (c) Evaluate the expression when x is 4:
`x += x++ * ++x % 2;`
- (d) Give the output of the following Character class methods:

- (i) `Character.toUpperCase('a')`
 - (ii) `Character.isLetterOrDigit('#')`
 - (e) Write the value of `n` after execution:
`char ch = 'd';`
`int n = ch + 5;`
 - (f) Give the output of the following:
 - (i) `Math.pow(36, 0.5) + Math.cbrt(125)`
 - (ii) `Math.ceil(4.2) + Math.floor(7.9)`
 - (g) Rewrite the following using ternary operator:
`if (n1 > n2)`
`r = true;`
`else`
`r = false;`
 - (h) Rewrite the following program segment using logical operators:
`if (x > 5)`
`if (x > y)`
`System.out.println(x + y);`
 - (i) Write the return data type of the following methods of `Character` class:
 - (i) `isLetter()`
 - (ii) `toUpperCase()`
 - (j) State the data type and value of `res` after the following is executed:
`char ch = '9';`
`res = Character.isDigit(ch);`
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Section-B

(60 Marks)

Attempt any four questions from Section B:

Q 5.

Write a program in Java that inputs a positive integer. Display all even numbers from 2 to that inputted positive integer, and display their sum also.

Q 6.

Write a program in Java that inputs the length and breadth of a rectangle. Calculate and display area, perimeter and diagonal with suitable message.

Q 7.

Write a program in Java that inputs the radius of a circle. Calculate and display area and perimeter of the circle with suitable message.

Q 8.

Write a program in Java that displays the first 100 natural numbers with suitable message. Display their sum also.

Q 9.

Write a program in Java that displays your name 50 times.

Q 10.

Write a program in Java that displays first 20 odd numbers with suitable message and display their sum also.